SOFTWARE DESIGN & ANALYSIS (SDA)

Desktop Application Project Proposals

Important Instructions for Students

- Each proposed system shall be implemented as a desktop application.
- The functional requirements given for the assigned project are fixed. Students are not allowed to remove, replace, modify, reinterpret, or reduce these requirements.
- Students shall not add web portals, mobile applications, online customer/self-service modules, cloud services, external APIs, e-mail/SMS services, GPS/live tracking, or online payment features unless explicitly stated in the assigned requirements.
- Users mentioned in the requirements interact with the system through the desktop application according to their assigned role.
- Students may make implementation-level assumptions only when necessary, provided that no functional requirement is changed.
- These requirements shall be used as the basis for project proposal, use cases, domain/class modeling, relationships, sequence diagrams, design, implementation, and testing.

PROJECT 1: LIBRARY CIRCULATION & INVENTORY MANAGEMENT SYSTEM

Application Type: Desktop Application

Requirement Status: Fixed — students shall implement all functional requirements listed below without changing them.

Member Management

FR1. The system shall allow the librarian to register a member by entering member ID, name, department, contact number, membership type and registration date.

FR2. The system shall distinguish Student, Faculty and Staff members and store the information required for each member type.

FR3. The system shall allow the librarian to search a member by member ID or name.

FR4. The system shall display a member's current borrowed items, overdue items, fines and previous borrowing history.

FR5. The system shall allow the librarian to update a member's contact information and membership status.

FR6. The system shall prevent an inactive member from borrowing an item.

Book and Copy Management

FR7. The system shall allow the librarian to register a book by ISBN, title, subject, edition, publisher and publication year.

FR8. The system shall allow one book title to have one or more authors.

FR9. The system shall maintain individual physical copies of a book using a unique copy number.

FR10. The system shall maintain the status of every copy as Available, Issued, Reserved, Lost or Damaged.

FR11. The system shall allow books to be classified into predefined categories.

FR12. The system shall allow the librarian to search books by title, ISBN, author or category.

Issue and Return

FR13. The system shall allow the librarian to issue an available book copy to an active member.

FR14. The system shall record issue date and due date for every issued copy.

FR15. The system shall enforce the borrowing limit defined for the member's membership type.

FR16. The system shall prevent the same physical copy from being issued to two members at the same time.

FR17. The system shall allow the librarian to receive a returned copy and record the return date.

FR18. The system shall automatically determine whether a returned item is overdue.

FR19. The system shall update the copy status to Available after a normal return.

Reservation and Fine

FR20. The system shall allow the librarian to place a reservation for a member when all copies of a requested book are unavailable.

FR21. The system shall maintain reservations in the order in which they are entered.

FR22. The system shall identify the next eligible member when a reserved book becomes available.

FR23. The system shall calculate an overdue fine using the fixed fine rules stored in the application.

FR24. The system shall allow the librarian to record full or partial fine payment.

FR25. The system shall maintain the outstanding fine balance for each member.

Lost/Damaged Items and Reports

FR26. The system shall allow the librarian to mark an issued copy as Lost or Damaged.

FR27. The system shall record a fixed replacement or damage charge against the responsible member.

FR28. The system shall provide a daily list of issued, returned and overdue books.

FR29. The system shall provide a list of books by category and availability status.

FR30. The system shall provide member-wise borrowing and fine reports.

FR31. The system shall provide a report of lost and damaged copies.

FR32. The system shall allow the librarian to print the selected report.

PROJECT 2: CLINIC RECEPTION & PATIENT RECORD MANAGEMENT SYSTEM

Application Type: Desktop Application

Requirement Status: Fixed — students shall implement all functional requirements listed below without changing them.

Patient Registration

FR1. The system shall allow the receptionist to register a patient using patient ID, name, date of birth, gender, address, contact number and emergency contact.

FR2. The system shall allow the receptionist to search an existing patient by patient ID, name or contact number.

FR3. The system shall allow authorized staff to update permitted patient information.

FR4. The system shall maintain a patient's previous visits and consultation records.

FR5. The system shall validate mandatory patient information before saving a new record.

Doctor Management

FR6. The system shall allow the administrator to register doctors with doctor ID, name, specialization, qualification and consultation fee.

FR7. The system shall allow a doctor to be associated with one department.

FR8. The system shall allow the administrator to define each doctor's working days and consultation time slots.

FR9. The system shall allow authorized staff to mark a doctor as Available or Unavailable for a specified day.

FR10. The system shall display the list of doctors by department and specialization.

Appointment Management

FR11. The system shall allow the receptionist to create an appointment for a registered patient with an available doctor.

FR12. The system shall record appointment date, time, token number and appointment status.

FR13. The system shall prevent two appointments from being assigned to the same doctor at the same date and time.

FR14. The system shall allow the receptionist to reschedule or cancel an appointment.

FR15. The system shall maintain appointment status as Scheduled, Completed, Cancelled or No-Show.

FR16. The system shall display a doctor's appointment list for a selected date.

Consultation and Prescription

FR17. The system shall allow the doctor to open the record of a patient scheduled with that doctor.

FR18. The system shall allow the doctor to record symptoms, diagnosis and consultation notes.

FR19. The system shall allow a doctor to create one prescription for a consultation.

FR20. The system shall allow a prescription to contain one or more medicines.

FR21. The system shall record medicine name, dosage, frequency and duration for each prescribed medicine.

FR22. The system shall store each completed consultation in the patient's medical history.

FR23. The system shall prevent unauthorized users from modifying a completed consultation.

Billing and Reports

FR24. The system shall create a bill for a completed consultation using the doctor's consultation fee and any fixed clinic service charges.

FR25. The system shall allow the receptionist to record payment and payment status.

FR26. The system shall generate a printable receipt for a paid bill.

FR27. The system shall provide daily and date-range appointment reports.

FR28. The system shall provide doctor-wise patient and consultation reports.

FR29. The system shall provide patient visit-history reports.

FR30. The system shall provide daily and date-range revenue reports.

PROJECT 3: VEHICLE RENTAL DESKTOP MANAGEMENT SYSTEM

Application Type: Desktop Application

Requirement Status: Fixed — students shall implement all functional requirements listed below without changing them.

Customer Management

FR1. The system shall allow rental staff to register a customer with customer ID, name, address, contact number, national ID and driving-license information.

FR2. The system shall allow rental staff to search a customer by customer ID, national ID or name.

FR3. The system shall store the customer's previous rental agreements.

FR4. The system shall allow authorized staff to update customer contact information.

FR5. The system shall prevent a customer with an expired driving license from receiving a vehicle.

Vehicle Management

FR6. The system shall allow fleet staff to register a vehicle with registration number, make, model, year, category, transmission type, fuel type, current mileage and daily rental rate.

FR7. The system shall classify vehicles into the fixed categories Economy, Sedan, SUV, Luxury and Van.

FR8. The system shall maintain vehicle status as Available, Reserved, Rented, Under Maintenance or Inactive.

FR9. The system shall allow authorized staff to search vehicles by registration number, category or status.

FR10. The system shall maintain the service and maintenance history of every vehicle.

Reservation

FR11. The system shall allow rental staff to create a reservation for a customer for a specified pickup date and return date.

FR12. The system shall display vehicles available for the requested rental period.

FR13. The system shall prevent overlapping confirmed reservations for the same vehicle.

FR14. The system shall calculate the estimated rental charge from the number of rental days and daily rental rate.

FR15. The system shall maintain reservation status as Pending, Confirmed, Cancelled, Collected or Expired.

FR16. The system shall allow rental staff to cancel a reservation before vehicle collection.

Rental Agreement and Return

FR17. The system shall allow staff to create a rental agreement from a confirmed reservation.

FR18. The system shall record pickup date/time, starting mileage, fuel level and existing damage at vehicle handover.

FR19. The system shall update the vehicle status to Rented after handover.

FR20. The system shall record actual return date/time, ending mileage, fuel level and newly observed damage.

FR21. The system shall calculate late-return, excess-mileage, fuel-shortage and damage charges using the fixed rules defined for the project.

FR22. The system shall calculate the final payable amount when the vehicle is returned.

FR23. The system shall update the vehicle status to Available after return unless maintenance is required.

Payment, Maintenance and Reports

FR24. The system shall allow staff to record payment against a rental agreement.

FR25. The system shall generate a printable rental invoice and payment receipt.

FR26. The system shall allow fleet staff to create a maintenance record containing maintenance type, date, description, cost and completion status.

FR27. The system shall prevent a vehicle with Under Maintenance status from being reserved or rented.

FR28. The system shall provide reports of Available, Reserved, Rented and Under Maintenance vehicles.

FR29. The system shall provide customer rental-history reports.

FR30. The system shall provide vehicle-wise rental and maintenance-history reports.

FR31. The system shall provide revenue reports for a selected date range.

PROJECT 4: HOSTEL ALLOCATION, FEE & COMPLAINT MANAGEMENT SYSTEM

Application Type: Desktop Application

Requirement Status: Fixed — students shall implement all functional requirements listed below without changing them.

Student and Hostel Management

FR1. The system shall allow hostel staff to register a student using student ID, name, program, semester, gender, contact number, guardian name and guardian contact.

FR2. The system shall allow hostel staff to register hostel buildings, floors and rooms.

FR3. The system shall store room number, room type, capacity, fee and room status.

FR4. The system shall maintain individual bed numbers for rooms having more than one bed.

FR5. The system shall display available beds by hostel, floor and room.

FR6. The system shall allow authorized staff to search student and room records.

Accommodation Application

FR7. The system shall allow hostel staff to enter an accommodation application received from a student.

FR8. The system shall record application date, requested room type and application status.

FR9. The system shall maintain application status as Submitted, Approved, Waiting List, Rejected or Cancelled.

FR10. The system shall allow the hostel manager to approve or reject an application.

FR11. The system shall display waiting-list applications in application-date order.

Room Allocation and Transfer

FR12. The system shall allow hostel staff to allocate one available bed to an approved student.

FR13. The system shall prevent one bed from being actively allocated to more than one student.

FR14. The system shall prevent a student from holding two active room allocations.

FR15. The system shall record allocation date and expected end date.

FR16. The system shall allow staff to record a student's room-transfer request with a reason.

FR17. The system shall allow the hostel manager to approve or reject a transfer request.

FR18. The system shall retain previous allocation information after a successful transfer.

Fee Management

FR19. The system shall generate the hostel fee for a student according to the allocated room type and fixed fee structure.

FR20. The system shall maintain charge types as Accommodation Fee, Security Deposit, Mess Charge and Fine.

FR21. The system shall allow accounts staff to record full or partial payment.

FR22. The system shall maintain each student's outstanding balance.

FR23. The system shall generate a printable receipt for every recorded payment.

FR24. The system shall display a student's fee ledger and payment history.

Complaint Management and Reports

FR25. The system shall allow hostel staff to register a complaint submitted by a resident.

FR26. The system shall classify complaints into Electrical, Plumbing, Furniture, Cleanliness, Internet and Security categories.

FR27. The system shall record complaint date, room, description, priority and status.

FR28. The system shall allow the complaint to be assigned to a maintenance employee.

FR29. The system shall maintain complaint status as Submitted, Assigned, In Progress, Resolved or Closed.

FR30. The system shall allow maintenance staff to record resolution details.

FR31. The system shall provide room and bed occupancy reports.

FR32. The system shall provide resident-allocation and waiting-list reports.

FR33. The system shall provide outstanding-fee and payment reports.

FR34. The system shall provide complaint reports by category, priority and status.

PROJECT 5: RETAIL STORE SALES & INVENTORY MANAGEMENT SYSTEM

Application Type: Desktop Application

Requirement Status: Fixed — students shall implement all functional requirements listed below without changing them.

Product and Supplier Management

FR1. The system shall allow the store manager to register products with product code, name, category, unit, selling price, reorder level and current status.

FR2. The system shall allow products to be classified into predefined product categories.

FR3. The system shall allow the store manager to register suppliers with supplier ID, name, address and contact information.

FR4. The system shall allow one supplier to supply multiple products and one product to be supplied by multiple suppliers.

FR5. The system shall record the supplier-specific purchase price for a product.

FR6. The system shall allow authorized users to search products by product code, name or category.

Purchase and Stock Receiving

FR7. The system shall allow the store manager to create a purchase order for one supplier.

FR8. The system shall allow a purchase order to contain one or more products with ordered quantity and purchase price.

FR9. The system shall calculate the total value of a purchase order.

FR10. The system shall maintain purchase-order status as Prepared, Ordered, Partially Received, Received or Cancelled.

FR11. The system shall allow store staff to record quantities received against a purchase order.

FR12. The system shall increase stock quantity only for quantities actually received.

FR13. The system shall maintain the date and quantity of each stock receipt.

Sales Transaction

FR14. The system shall allow a cashier to create a new sales transaction.

FR15. The system shall allow one sale to contain one or more products with quantity and selling price.

FR16. The system shall display the current stock quantity before a product is added to a sale.

FR17. The system shall prevent the cashier from selling a quantity greater than available stock.

FR18. The system shall calculate line totals, sale subtotal, fixed applicable discount and final amount.

FR19. The system shall reduce product stock after the sale is finalized.

FR20. The system shall record sale date/time and responsible cashier.

FR21. The system shall generate a printable sales receipt.

Returns and Stock Adjustment

FR22. The system shall allow the cashier to record a customer return only against an existing completed sale.

FR23. The system shall record returned product, quantity, reason and return date.

FR24. The system shall update stock for accepted reusable returned items.

FR25. The system shall allow authorized store staff to record stock adjustments for damaged, expired or missing products.

FR26. The system shall record adjustment type, quantity, date and reason.

FR27. The system shall maintain a history of all stock adjustments.

Users and Reports

FR28. The system shall distinguish Manager, Cashier and Storekeeper users.

FR29. The system shall require a valid username and password before allowing access to the application.

FR30. The system shall restrict functions according to the user's role.

FR31. The system shall provide a low-stock report for products at or below their reorder level.

FR32. The system shall provide current inventory and stock-movement reports.

FR33. The system shall provide daily and date-range sales reports.

FR34. The system shall provide supplier-wise purchase reports.

FR35. The system shall provide product-wise sales reports.

FR36. The system shall allow authorized users to print selected reports.